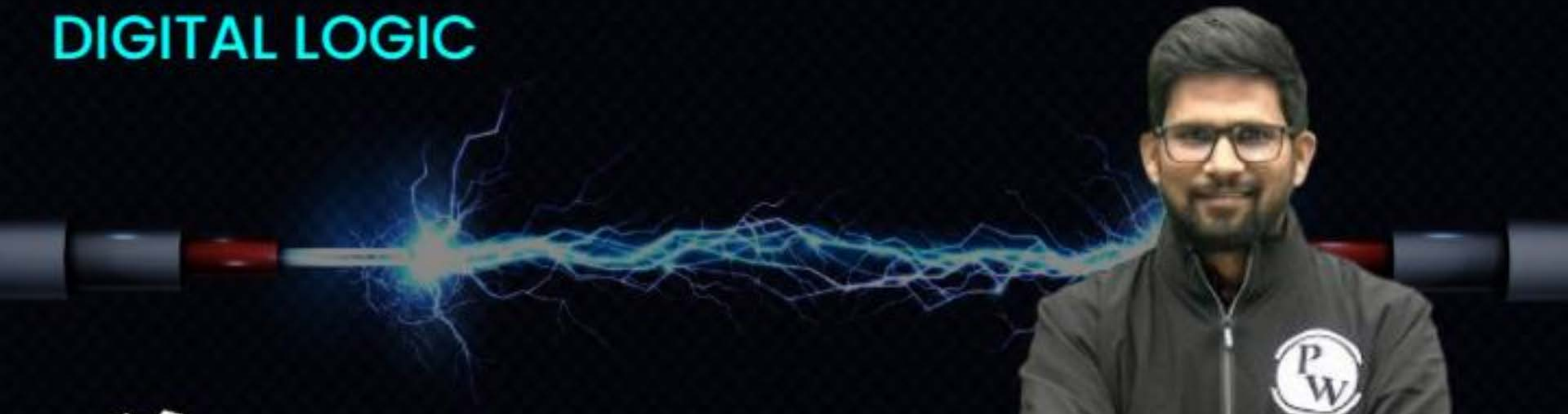
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 02

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- S-R latch
- S-R FF
-



S-R F F

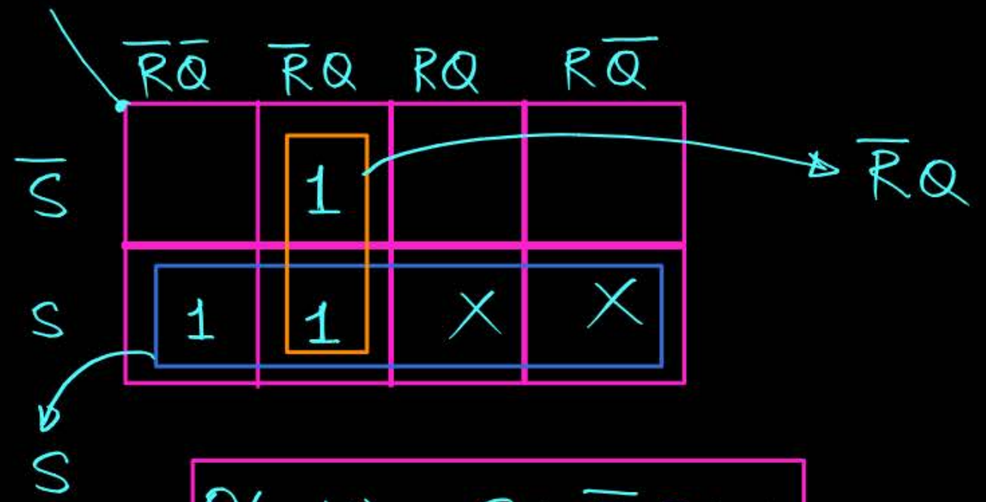
J-K-FF

Characteristic Table of SR Flip-Flop

Clock active

S	R	$Q(n)$	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

$$Q[n+1][S, R, Q(n)] = \Sigma(1,4,5) + d\Sigma(6,7)$$

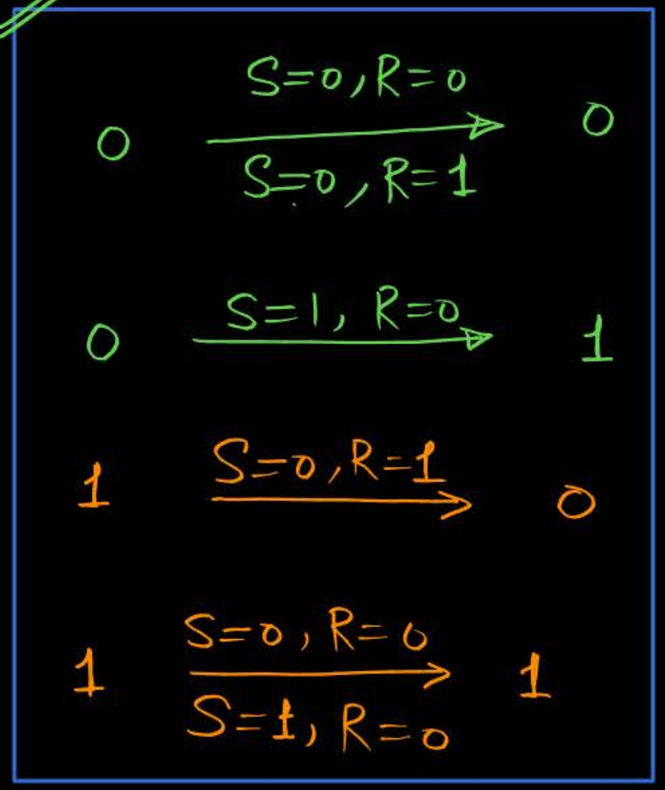


$$Q(n+1) = S + \bar{R}\,Q(n)$$

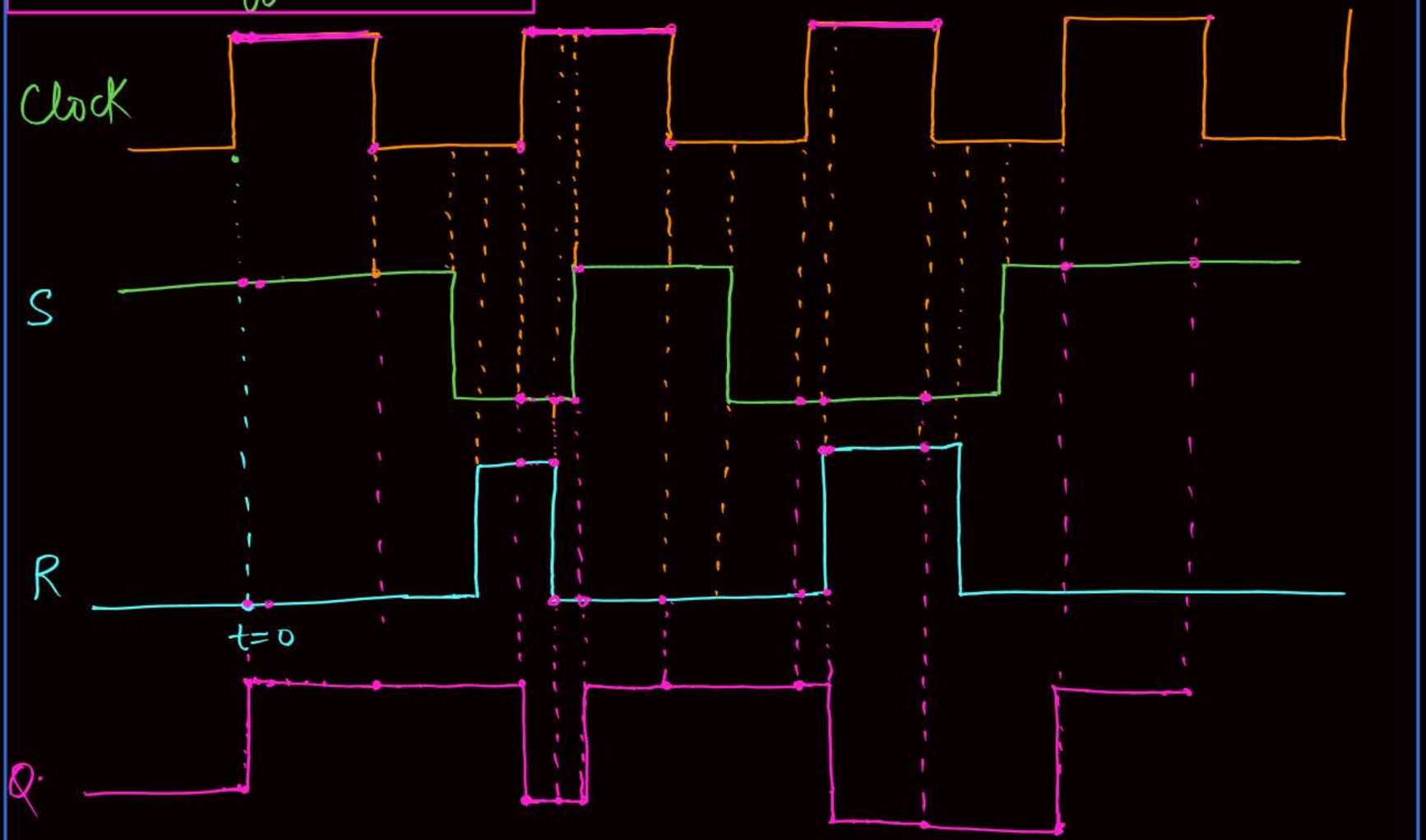
Valid when $S \cdot R = 0$

Excitation Table

$Q(n)$	$Q(n+1)$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



+ve level triggered S-R FF



H.W.

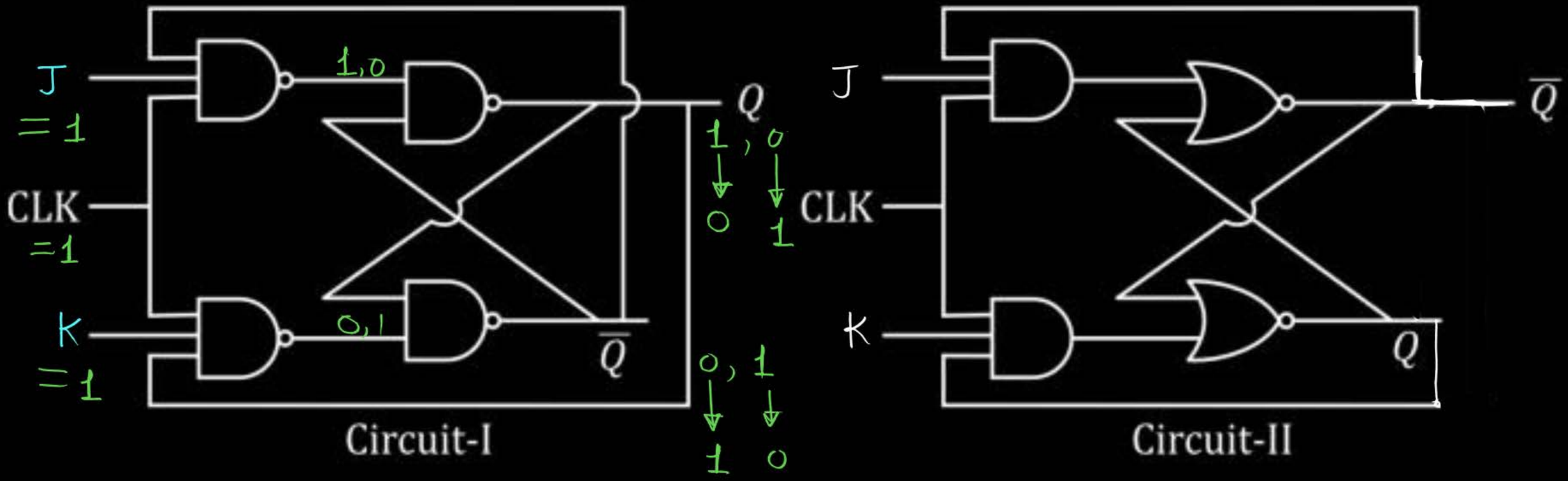
if ff is -ve level triggered then draw the O/p wave form.

- Problem with S-R flip-flop :

- S-R ff generates invalid o/p corresponding to S=1, R=1 and that could be problematic in circuit design.
  → to avoid this problem J-K ff is designed.

J-K Flip-Flop

- Designed to avoid invalid state of S-R Flip-Flop.
- Circuit :



- Working :

	J	K	Q(n+1)
Not active	X	X	Q(n)
Clock active	0	0	Q(n) ⟶ hold state
"	0	1	0
"	1	0	1
"	1	1	$\overline{Q}(n)$ ⟶ toggle state

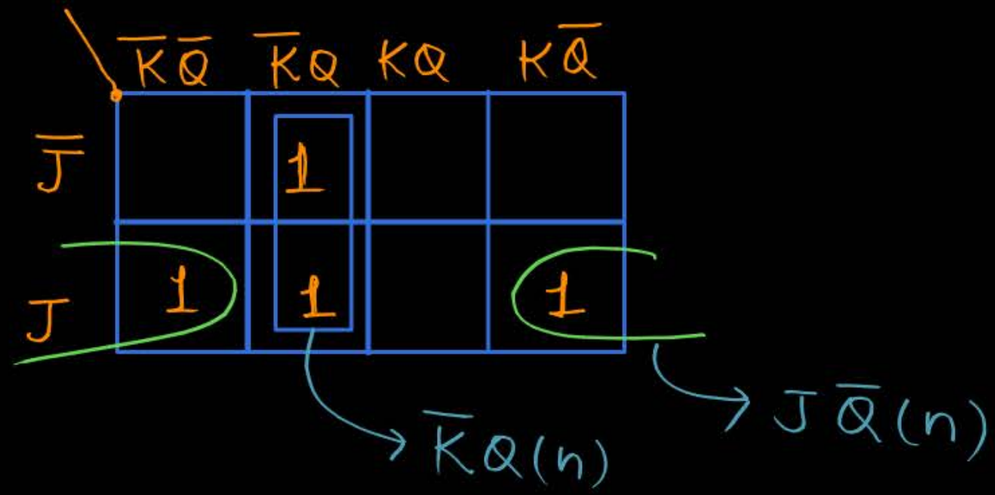
Characteristic Table :

Clock-active

J	K	$Q(n)$	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$Q[n+1][J, K, Q(n)]$

$= \Sigma(1,4,5,6)$



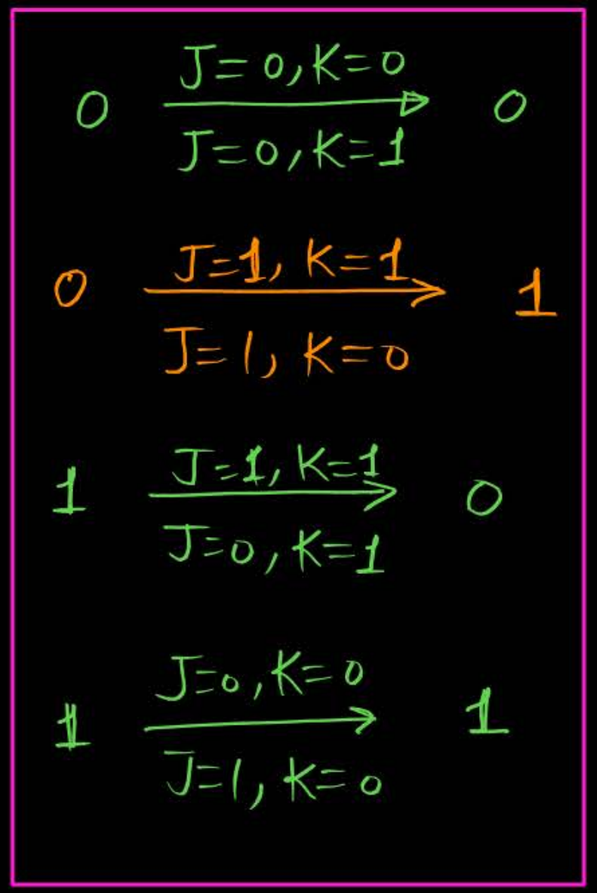
$$Q(n+1) = J\bar{Q}(n) + \bar{K}Q(n)$$

→ always valid.

Excitation Table

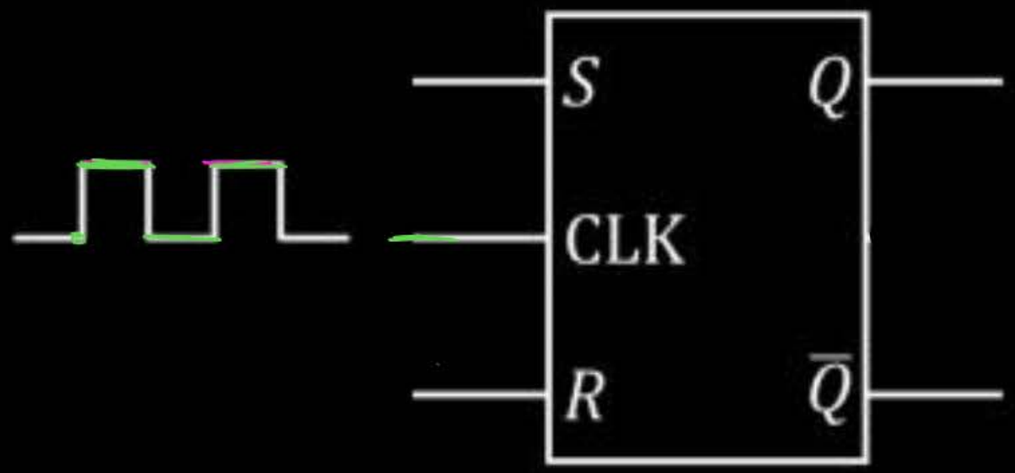
$Q(n)$	$Q(n+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



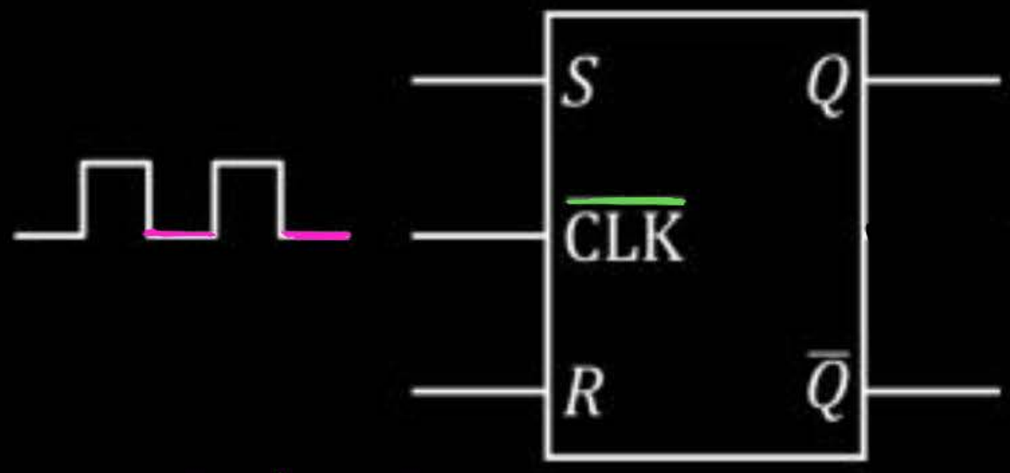


Level Trigger S-R Flip-Flop

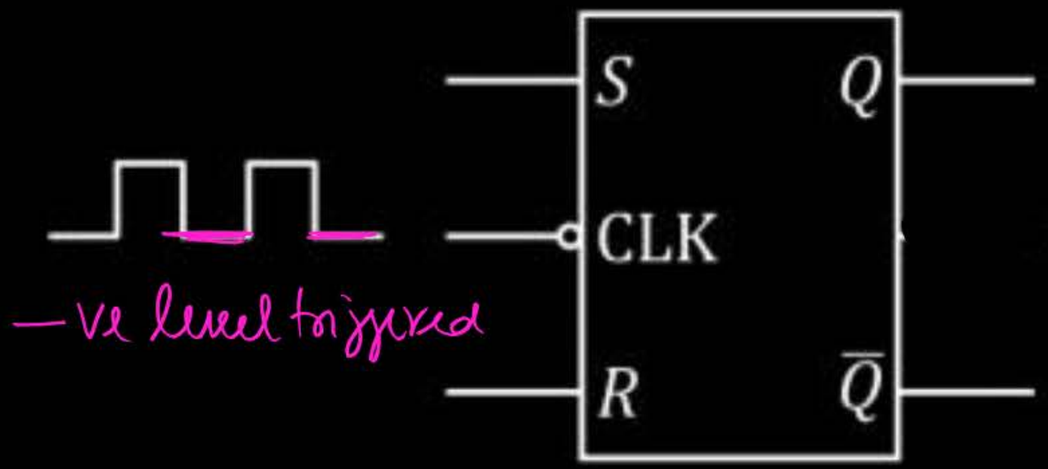
Different representation of level triggered FF's



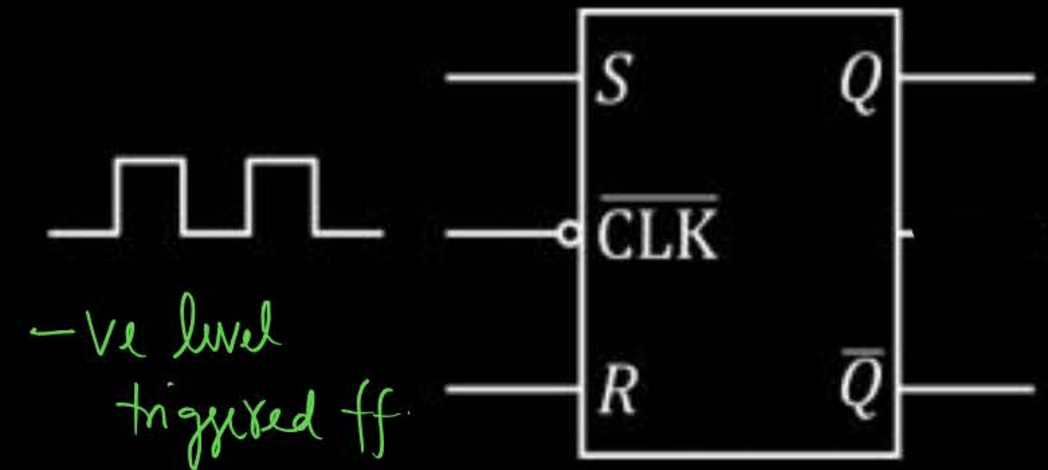
+ve level triggered



−ve level triggered

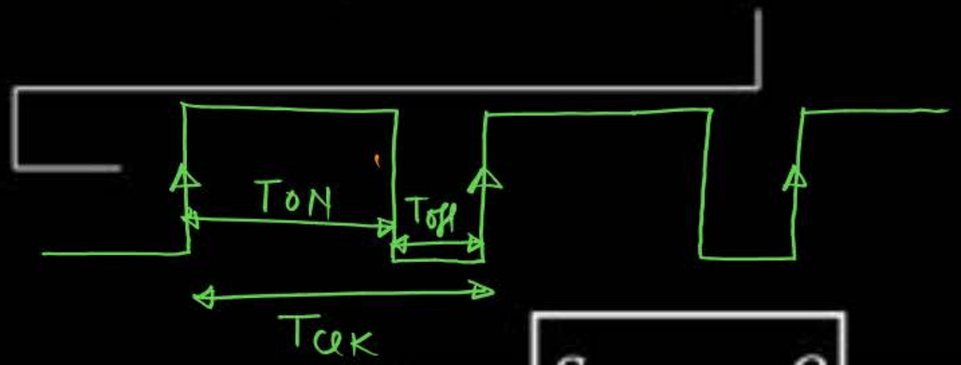


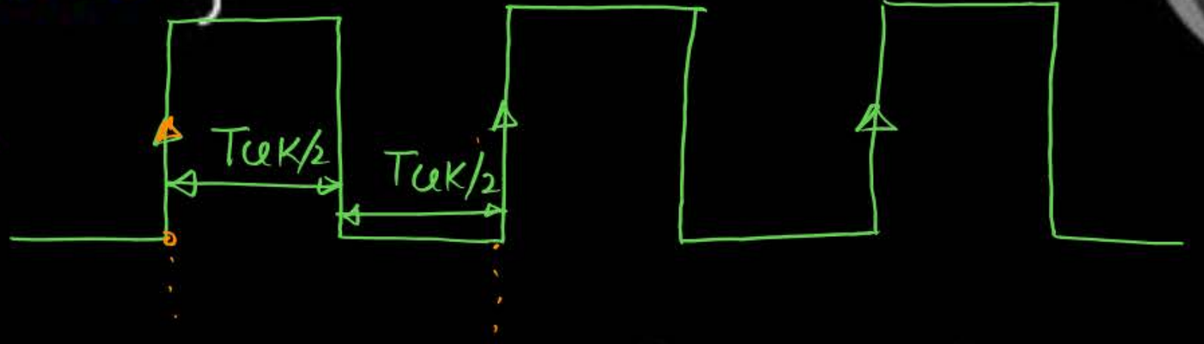
-ve level triggered

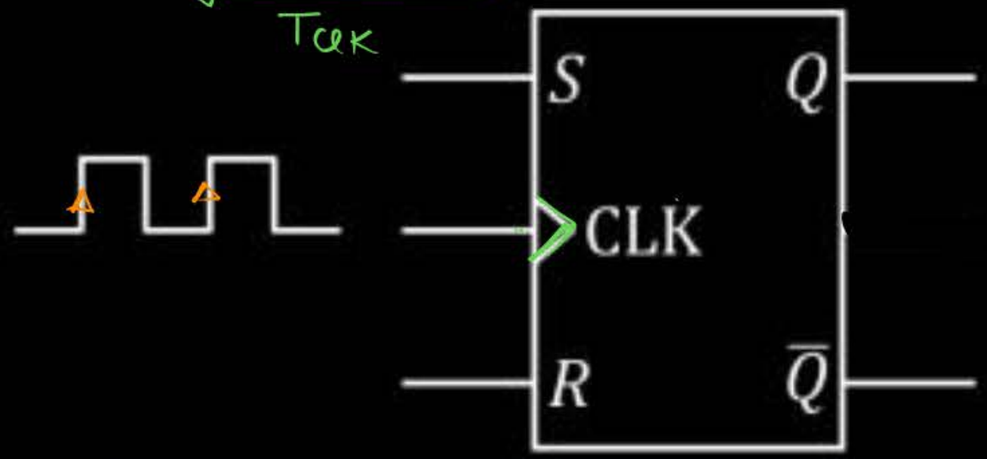


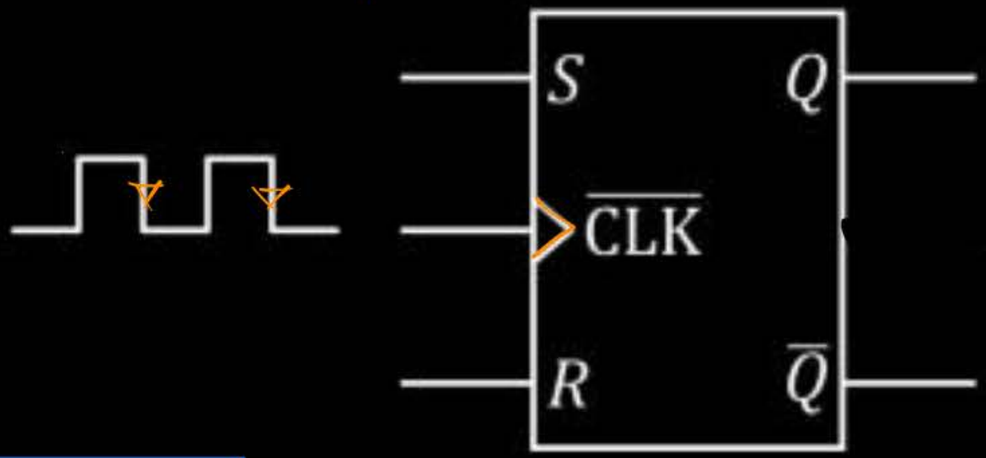
-ve level triggered ff.

Edge Triggered FF



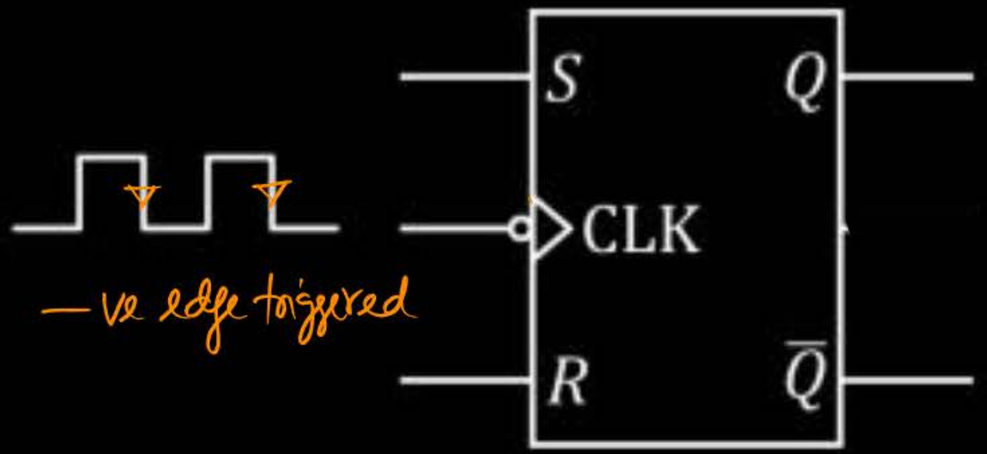




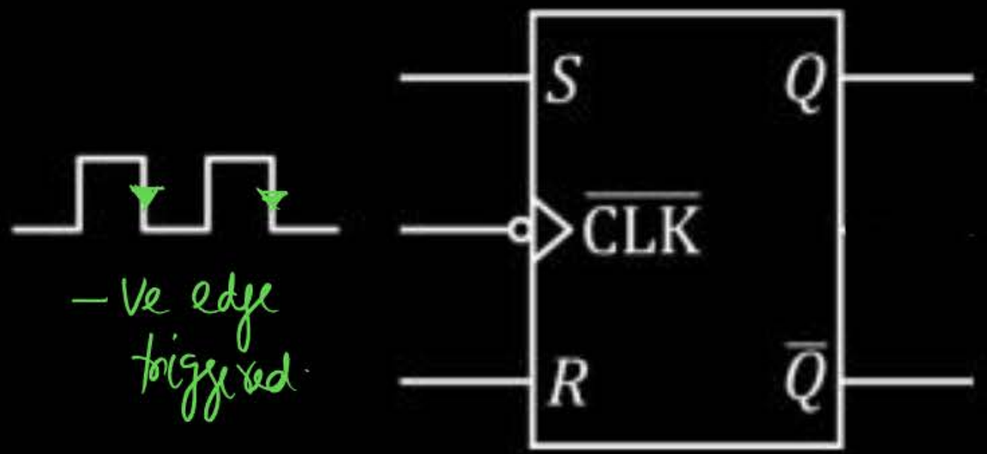


+Ve edge triggered → triggered when clock take transition from (0–1)

–Ve edge triggered (triggered when clock take transition from (1–0)).



– ve edge triggered



– ve edge triggered.

- IMP points about level triggered and edge triggered :

- In case of level triggered ff O/P may change more than once in a single clock.
- While in case of edge triggered ff O/P changes only once in a single clock.



Topic : 2 Min Summary

- S-R ff
- J-K ff
- triggering.

Thank you

GW

Soldiers !

